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Intrusion Detection System Development Making Use of Machine Learning

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Abstract - The escalating frequency of network attacks poses a significant threat to the confidentiality, availability, and integrity of sensitive data for both individuals and organizations. This paper proposes an intrusion detection approach grounded in supervised machine learning. The methodology is both simple and effective, offering flexibility for integration with various machine learning models. To evaluate the efficiency of the proposed Intrusion Detection System (IDS), several well-known machine learning algorithms are tested. Experimental results reveal that Support Vector Machines (SVM) and K-Nearest Neighbor (KNN) algorithms achieve superior performance compared to other techniques. Based on these findings, an IDS is constructed using SVM and KNN to classify network traffic as either legitimate or malicious. Additionally, key network data features crucial for attack detection are identified using information gain for feature selection. The system demonstrates a detection accuracy exceeding 98% for distinguishing normal behavior from major attack categories such as Probe and Denial of Service (DoS), all within a minimal processing window. To further enhance the system's performance, a novel post-processing technique is implemented, significantly reducing false alarms and improving the overall accuracy and reliability of the IDS.

Key Words: Intrusion Detection System, Support Vector Machine, K-Nearest Neighbor, LCCDE, Machine Learning, Ensemble Framework, Feature Selection

I. INTRODUCTION

Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS) are among the most critical tools for defending against the growing complexity and sophistication of modern cyber-attacks [1] [8]. Anomaly-based intrusion detection methods are continuously evolving in accuracy and reliability; however, their development is often hindered by the lack of comprehensive and trustworthy datasets for testing and validation [13]. Offline intrusion detection systems function by routinely analyzing stored network logs or historical traffic data to detect abnormal or malicious activities. In contrast, online IDS solutions operate in real time, leveraging key features such as timestamps, IP addresses (source and destination), port numbers, protocols, and attack indicators [11] (typically stored in CSV formats) to monitor ongoing traffic and detect threats as they occur. With the rapid integration of internet services into both personal and business domains, the dependency on network infrastructure has surged bringing with it increased exposure to remote attacks and data breaches. To secure sensitive information and digital assets, organizations must implement robust security measures, which may include firewalls, antivirus tools, and IDS technologies as part of their network architecture. However, defending a corporate network from a diverse range of threats requires more than just a firewall [12]. Firewalls often fall short when it comes to blocking attacks that exploit open ports which are essential for regular network communication. Due to this limitation, intrusion detection systems are commonly deployed alongside firewalls. These systems gather data from computers or network components and analyze it for indicators of unauthorized or suspicious activity. Network intrusion detection systems generally fall into two main types: host-based and network-based. Host-based IDS directly monitors and analyzes data from the specific system being protected. In contrast, network-based IDS functions at the perimeter—such as at a server or gateway—where it inspects live network traffic before it can reach internal systems. Moreover, IDS technologies can operate in either online (real-time) or offline (post-event) detection modes [11], providing flexible options for security monitoring and threat response.

KDD99 dataset is widely recognized for containing both normal (benign) and commonly occurring types of network attacks, closely mimicking real-world scenarios [13] captured through packet capture files (PCAPs). This dataset also incorporates flow-level data generated using CIC Flow Meter, which labels network traffic based on parameters such as timestamps, IP addresses (source and destination), ports, communication protocols, and identified attack types. These flows are provided in structured CSV format, and a clear description of the extracted

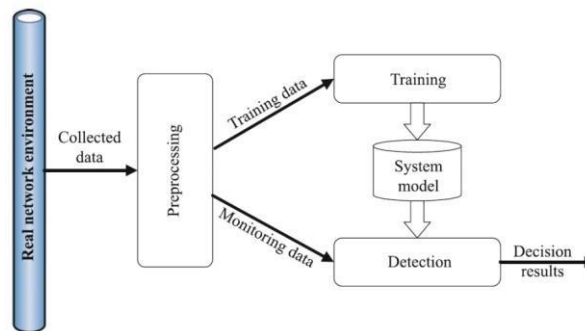


Figure 1: Block Diagram

features is also available. This study addresses both network-based and host-based intrusion detection. It captures live network traffic and processes it in real time, allowing the system to detect threats and promptly notify administrators, enabling immediate intervention to stop ongoing attacks. The proposed detection framework can also be deployed effectively at the host level, offering flexible defense capabilities. Our IDS leverages a misuse detection approach, which is designed to classify attacks into specific categories. Unlike anomaly detection, which simply flags deviations from normal behavior, misuse detection provides more detailed insight into the type of intrusion. Although a wide range of network features can be analyzed for intrusion detection, this work focuses on a subset of 12 key features extracted from packet headers. These carefully chosen attributes have proven effective in distinguishing normal behavior from malicious activities. Specifically, they enable accurate classification of major attack types such as Port Scanning (Probing) and Denial of Service (DoS) attacks. By narrowing the feature set, we reduce computational complexity, leading to faster detection rates and lower resource usage in terms of CPU and memory. Block diagram is illustrated in figure 1.

II. RELATED WORK

Previous studies in the field have predominantly focused on offline intrusion detection techniques, primarily utilizing the KDD99 benchmark dataset, which has been widely accepted for evaluating IDS performance since its release by DARPA in 1999 [13]. Although there are some methods that explore online detection, offline approaches remain more common. Given the critical importance of protecting sensitive digital assets in both organizational and individual contexts, the role of Intrusion Detection Systems (IDS) has become increasingly vital. A breach in security could lead to severe consequences, including financial and data losses. To enhance the capabilities of IDS, numerous machine learning algorithms have been employed. One such study [2] aimed to tackle the adaptability issue in IDS, proposing a system capable of identifying both known and previously unseen threats. The model integrates three key components: a Clustering Manager (CM) for grouping similar behaviors, a Decision Maker (DM) for classifying activities, and an Update Manager (UM) for system refinement. The effectiveness of this hybrid system was evaluated using the CICIDS2017 dataset [2], employing both supervised and unsupervised learning paradigms to optimize adaptability. Experimental results suggest that the system is capable of detecting intrusions by learning patterns from both labeled and unlabeled data. In supervised mode, the agent is trained to ignore misleading feedback, while the unsupervised mode enables the detection of unfamiliar attack patterns. The system updates itself continually using newly encountered data, which further enhances its performance over time. Another research initiative developed a hybrid detection model that incorporates Support Vector Machines (SVM) and Extreme Learning Machine (ELM) algorithms [3][7]. A modified K-means clustering technique was introduced to create a compact, high-quality training set representative of the original data. This strategy significantly reduced the training time without compromising accuracy. Using the KDDCUP 1999 dataset, the model achieved an accuracy rate of approximately 95.75%. Comparisons among SVM, Random Forest (RF), and ELM revealed that while SVM performed well with smaller sample sizes, ELM outperformed others when handling large-scale data, particularly datasets exceeding 200,000 instances [7]. With the rise of interconnected devices and an ever-growing digital infrastructure, cybersecurity threats have become increasingly sophisticated. Intrusion Detection and Prevention Systems (IDPS) must evolve to detect emerging threats and respond effectively. IDS architecture generally consists of three primary stages: data collection, feature selection or transformation, and a decision-making engine. Machine learning offers valuable tools to automate model training for IDS. As a subset of artificial intelligence, it enables systems to recognize patterns and make decisions with limited human input. Supervised and unsupervised learning are the most prevalent methodologies. Supervised learning leverages labeled datasets for training, while unsupervised learning identifies latent structures within unlabeled data. Additionally, semi-supervised and reinforcement learning techniques are used in more advanced scenarios. Semi-supervised learning combines small labeled datasets with a large volume of unlabeled data, while reinforcement learning allows an agent to improve decision-making through feedback-driven interaction with its environment. To further enhance IDS effectiveness, correlation-based feature selection (CFS) has been used to filter features that are strongly linked to the

target class but minimally redundant with one another. This approach was applied using the CICIDS2019 and UNSW-NB15 datasets, achieving a 99% detection rate and a remarkably low 0.01% false positive rate [4]. Additionally, a hybrid intrusion detection framework was proposed, integrating AdaBoost and Artificial Bee Colony (ABC) algorithms to maximize detection accuracy while minimizing false positives [4].

III. PROPOSED METHODOLOGY

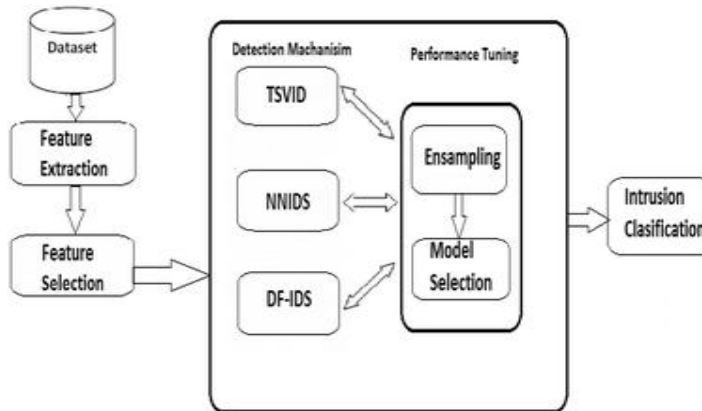


Figure 2: System Architecture

Machine Learning Approaches for Intrusion Detection

Intrusion detection is essentially a classification task, where the system is required to distinguish between legitimate and malicious network activity. This process relies on extracting meaningful features from the network traffic data. Selecting the most effective subset of these features is critical [6], as it significantly affects both processing efficiency and system performance. Reducing the feature set to only those that are most relevant helps lower computational overhead and resource usage. Machine Learning (ML), a subfield of Artificial Intelligence (AI), automates the development of predictive models by learning patterns from data. It enables systems to improve their decision-making capabilities with minimal human oversight. ML primarily employs two types of learning strategies: supervised learning [10], where models are trained using labeled datasets, and unsupervised learning, where the system identifies structures and relationships within data without predefined labels.

Other useful ML approaches include semi-supervised learning, which combines a small set of labeled data with a larger pool of unlabeled data to enhance training efficiency, and reinforcement learning, which is based on an interactive feedback loop [6]. In reinforcement learning [6] [17], an intelligent agent operates within an environment and chooses actions that yield optimal cumulative rewards over time. The main components involved are the agent, the environment, and the available actions. The goal is to establish a decision-making strategy or policy that results in the most beneficial outcomes.

Intrusion Detection Process and Algorithms

This section covers the evaluation process and the experimental results of the proposed IDS system. It begins with an overview of the network data used for testing, followed by the design of the experiments and the performance metrics applied. The final part discusses the results and findings of the detection system. Intrusions can severely compromise both the data and the physical components of a computer network [5] [18]. The urgency of timely detection lies in the fact that attackers can cause substantial damage within seconds. While several detection techniques exist, achieving high precision remains a key issue. Performance is generally measured by two factors: the rate of correctly identified intrusions and the rate of false positives [5] [13]. A system must strike a balance between detecting threats effectively and minimizing unnecessary alerts. To improve classification accuracy, machine learning models like Support Vector Machines (SVM) and Naïve Bayes classifiers are utilized [5]. Alongside these, data normalization and feature reduction techniques are applied to refine the input data and facilitate a comparative study of performance. A new hybrid model that integrates the Artificial Bee Colony (ABC) algorithm with the Artificial Fish Swarm (AFS) algorithm has also been introduced to enhance classification capability [4]. As internet connectivity expands, so does the potential for security breaches, making IDS technologies more essential than ever. Fuzzy C-Means (FCM) clustering is implemented for segmenting the training dataset, while Correlation-Based Feature Selection (CFS) helps eliminate features that add little value. To create decision rules that separate regular behavior from anomalies, the CART (Classification and Regression Tree) technique is employed [4]. During the threat detection phase, a deep learning-based model is trained to identify suspicious behavior within the network. This is achieved through a sequence of model-building steps aimed at maximizing predictive accuracy while minimizing error. In the system architecture, a designated part of the network functions as a data handling node. Any transmission to other nodes is first routed through a central server, which then forwards the data ensuring it is monitored and secured before reaching its destination. System architecture for proposed architecture is shown in figure 2.

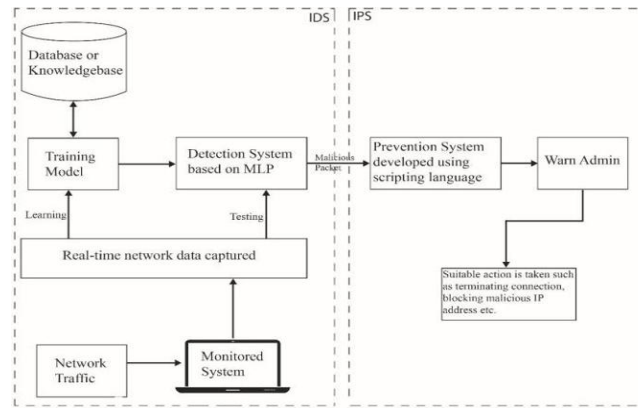


Figure 3: Intrusion Detection and Prevention System Architecture

i. Support Vector Machine (SVM)

Support Vector Machine (SVM) is a powerful algorithm primarily used for classification (figure 4) tasks in supervised learning. Its key objective is to identify the most suitable boundary called a hyperplane that distinctly separates data into different categories in a multi-dimensional space. This boundary is built using a subset of data points, known as support vectors, which are the most influential in determining the margin between the classes. These support vectors help the model generalize well, making it capable of correctly classifying unseen inputs [5] [7].

ii. K-Nearest Neighbor (KNN)

K-Nearest Neighbor (KNN) is among the simplest yet highly effective supervised classification (figure 5) algorithms. Unlike some models that assume specific distributions (e.g., Gaussian), KNN is non-parametric, meaning it doesn't rely on assumptions about how the data is spread. It classifies new instances by measuring their similarity (usually by distance) to known data points in the training set. The category is determined based on the majority class among the 'k' closest neighbors [5]. KNN is widely adopted in tasks such as anomaly detection, pattern recognition, and security systems. Interestingly, no single model performs best across all types of threats while one model may accurately detect DoS attacks, another might excel in identifying probing or sniffing attempts.

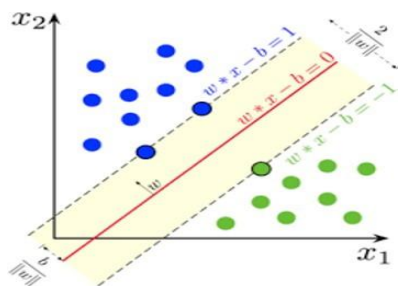


Figure 4: SVM Classifier

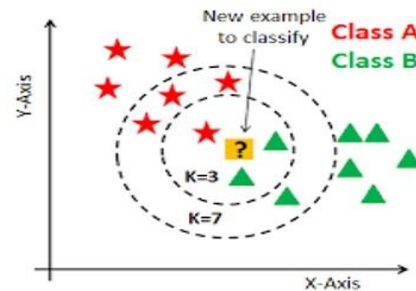


Figure 5: KNN Classifier

iii. Proposed Ensemble Model (LCCDE-Leader Class and Confidence Decision Ensemble (LCCDE))

To enhance accuracy over a broad range of attack types, an ensemble approach is proposed [20]. Rather than relying on a single learning model, this method combines the predictive strength of multiple classifiers. The ensemble incorporates three high-performance gradient boosting algorithms XGBoost, LightGBM, and CatBoost [15] [16]. This integration improves detection reliability by leveraging each algorithm's strengths, making the overall system more adaptable and precise in identifying network intrusions. A detail flow is shown in figure 3 [8] [9].

IV. RESULT AND DISCUSSIONS

TABLE 1
ACCURACY COMPARISON BETWEEN DIFFERENT ALGORITHMS

Algorithms	Accuracy
LCCDE	99.42 %
SVM	98.24%
KNN	98.02%

The proposed LCCDE model observes an improved detection accuracy of 99.42%. Table 1 shows accuracy values compared between different algorithms. Graphical comparison is shown in figure 6.

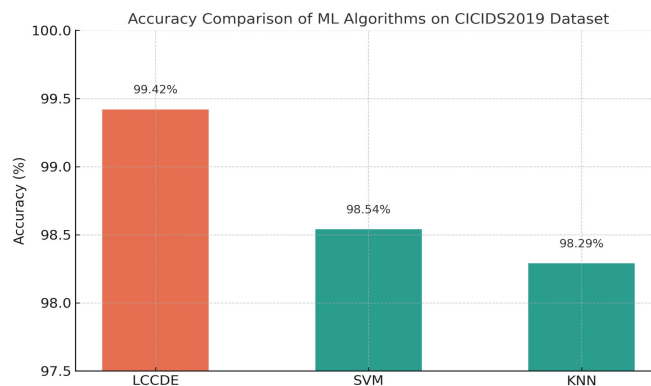


Figure 6: Graphical Comparison for Various Algorithms

V. CONCLUSION AND FUTURE SCOPE

Machine learning (ML) techniques have been employed as promising ways for recognizing different kinds of cyberattacks in order to improve IoT network security [14] [18] [19]. However, the performance of ML models varies depending on the type of attack detection. Nevertheless, ML models frequently exhibit varying performance for distinct attack detection types. This research proposes a unique ensemble method called Leader Class and Confidence Decision Ensemble (LCCDE) to achieve optimal performance on all forms of attack detection in IoT networks. In order to build a strong ensemble model, it selects the top-performing machine learning models for every kind of attack detection as the leader class models. The final prediction classes are also determined with the use of the prediction confidence data. The suggested LCCDE ensemble model is built using three sophisticated gradient boosting machine learning algorithms XGBoost, LightGBM, and CatBoost because of their great efficacy and efficiency. High accuracy of 99.42% is obtained by the suggested IDS framework through the experiments on CICIDS2019 datasets. Furthermore, the F1-scores of the suggested model outperform those of other examined machine learning techniques for identifying all attack types. This demonstrates the advantages of the suggested leader class-based approach.

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